

# **SALES PERFORMANCE ANALYSIS AND PROFITABILITY ASSESSMENT FOR NIKE RETAIL SPORTSWEAR SEGMENT**

## **Executive Summary**

This project analyzed sales performance, revenue generation, profitability trends, customer segments, product performance, regional performance, and sales channel effectiveness of Nike Retail Sportswear Segment. The objective was to identify key business drivers, evaluate profitability, and provide data-driven recommendations for improving business performance.

The analysis revealed that the Training product line generated the highest revenue and profit, Bengaluru emerged as the most profitable region, and the Online sales channel outperformed Retail sales. Seasonal trends showed peak revenue generation in December, while analysis of loss-making transactions identified opportunities for improving pricing, discount management, and operational efficiency.

Overall, the business demonstrated strong profitability across product lines and regions, with clear opportunities for revenue growth and profit optimization.

## **1. Introduction**

Data-driven decision-making has become increasingly important in today's competitive retail environment. Organizations rely on sales analytics to understand customer behavior, monitor business performance, identify growth opportunities, and improve profitability.

This project applies data analytics techniques to evaluate sales performance of Nike sportswear business segment operating across multiple regions in India. The analysis focuses on revenue generation, profitability, customer segments, product performance, and regional trends.

## **2. Business Objectives**

The primary objectives of this analysis were to:

- Evaluate overall sales and profitability performance.
- Identify the highest-performing product lines and products.
- Assess regional sales and profit performance.
- Analyze customer purchasing behavior by gender category.
- Evaluate the effectiveness of sales channels.
- Identify seasonal revenue patterns.
- Investigate loss-making transactions and operational risks.
- Generate actionable business recommendations.

## **3. Dataset Overview**

The dataset contains transactional sales records across multiple regions, product categories, customer segments, and sales channels.

Key variables included:

- Order Date
- Product Name
- Product Line
- Revenue
- Profit
- Units Sold
- Region
- Gender Category
- Sales Channel
- Discount Applied

The dataset was used to assess business performance over the period 2023–2025.

## **4. Data Quality Assessment and Cleaning**

### **Overview**

Prior to analysis, a comprehensive data quality assessment was conducted to evaluate the completeness, consistency, and reliability of the dataset.

Several data quality issues were identified and addressed before conducting exploratory analysis.

### **Data Quality Issues Identified**

The following issues were observed:

- Missing values in Order Date, Units Sold, Size, and Discount Applied.
- Negative values in Revenue, Profit, and Units Sold.
- Inconsistent regional naming conventions.
- Date formatting inconsistencies.

### **Column-Level Data Cleaning Approach**

Given the relatively small size of the dataset, a column-level data cleaning approach was adopted. Each variable was individually assessed to identify issues specific to that field and determine the most appropriate treatment method.

This approach ensured that cleaning decisions remained aligned with business context while preserving the integrity of the data.

### **Missing Value Treatment**

- Missing Units Sold values were treated using median imputation.
- Missing Size values were replaced using the modal category.
- Missing Order Date values were retained but excluded from date-based analyses.
- Missing Discount Applied values were reviewed and treated appropriately.

### **Category Standardization**

Regional naming inconsistencies were corrected to improve reporting accuracy. For example, duplicate representations of the same city were standardized into a single category.

### **Investigation of Negative Values**

Negative values were investigated rather than automatically removed.

The analysis identified:

- 578 negative-profit transactions.
- 53 negative-revenue transactions.
- 205 negative Units Sold records.

These records were retained where appropriate because they may represent product returns, refunds, inventory adjustments, or pricing-related issues.

### **Data Validation**

Following cleaning, validation checks were performed to confirm:

- Correct data types.
- Successful treatment of missing values.
- Consistency of categorical variables.
- Accuracy of derived Year and Month fields.

The cleaned dataset was subsequently used for exploratory data analysis.

## **5. Exploratory Data Analysis**

### **Revenue by Product Line**

Training generated the highest revenue at ₹195,552.89, followed by Basketball (₹155,292.42) and Lifestyle (₹148,870.70). Running generated the lowest revenue at ₹72,075.95.

**Insight:** Training products represent the strongest revenue-generating category and should remain a strategic focus for marketing and inventory planning.

### **Profit by Product Line**

Training generated the highest profit at ₹737,669.90, followed by Lifestyle (₹694,085.85) and Basketball (₹676,638.51).

**Insight:** All product lines contributed positively to profitability, indicating a balanced and healthy product portfolio.

### Revenue by Region

Kolkata generated the highest revenue at ₹168,600.77, followed by Hyderabad and Bengaluru.

**Insight:** Kolkata demonstrated the strongest sales performance and represents a key revenue-generating market.

### Profit by Region

Bengaluru generated the highest profit at ₹621,948.74, followed by Kolkata (₹602,856.74) and Delhi (₹587,148.93).

**Insight:** Bengaluru emerged as the most profitable region, reflecting strong operational performance and market demand.

### Revenue by Sales Channel

Online sales generated ₹378,183.10, while Retail sales generated ₹309,001.19.

**Insight:** Customers demonstrated a stronger preference for online purchasing channels, highlighting the importance of digital sales strategies.

### Monthly Revenue Trend

Revenue fluctuated throughout the year, with December generating the highest revenue of ₹73,355.11 and February recording the lowest revenue of ₹25,250.83.

**Insight:** Revenue exhibits clear seasonality, with demand peaking during year-end periods.

### Yearly Profit Trend by Region

Profitability increased significantly between 2023 and 2025 across all regions.

Kolkata recorded the highest regional profit in 2024 at ₹238,264.83, while Bengaluru emerged as the strongest performer in 2025 with ₹226,004.82.

Although profitability remained strong overall, Hyderabad and Kolkata experienced declines in 2025 compared to 2024.

**Insight:** The business experienced substantial growth, although certain regions may be facing increasing operational or market pressures.

## Top Profit-Making Products

The highest profit-generating products were:

1. LeBron 20 – ₹220,287.24
2. SuperRep Go – ₹212,719.79
3. Flex Trainer – ₹199,011.40

**Insight:** A small group of products contributes significantly to overall profitability and should remain priority products.

## Loss-Making Products (Transaction-Level Losses)

Products associated with the largest transaction-level losses included:

- Free RN (-₹21,277.95)
- Air Jordan (-₹20,834.52)
- Waffle One (-₹20,519.42)

**Insight:** These products remain profitable overall but experienced significant losses in certain transactions, warranting review of pricing and discount practices.

## Revenue by Gender Category

Men's products generated the highest revenue at ₹261,454.48, followed by Women's products at ₹232,600.58 and Kids' products at ₹193,129.23.

**Insight:** Men's products represent the strongest customer segment and offer significant opportunities for continued growth.

## 6. Key Findings

The analysis identified several important findings:

- Training is the highest-performing product line in both revenue and profit.
- Bengaluru is the most profitable region.
- Kolkata generates the highest regional revenue.
- Online sales outperform Retail sales.
- December is the strongest sales month.
- LeBron 20 is the most profitable product.
- Men's products generate the highest revenue among customer segments.
- Loss-making transactions remain a notable operational concern.

## 7. Recommendations

Based on the findings, the following recommendations are proposed:

1. Increase investment in high-performing product lines such as Training and Basketball.
2. Expand digital marketing initiatives to capitalize on the strong performance of Online sales channels.
3. Investigate products associated with recurring transaction-level losses to identify pricing or discount-related issues.
4. Leverage seasonal demand by increasing inventory availability ahead of peak sales periods, particularly December.
5. Conduct further analysis of Hyderabad and Kolkata to understand the causes of declining profitability in 2025.
6. Develop targeted campaigns for Women and Kids product categories to diversify revenue sources.
7. Strengthen data governance practices to minimize missing values and improve data quality.

## **8. Conclusion**

This project demonstrated the application of data cleaning, exploratory data analysis, and business intelligence techniques to evaluate Nike retail sales performance.

The analysis revealed strong overall business performance driven by Training products, Bengaluru, Online sales channels, and the Men's customer segment. While profitability increased substantially across the study period, opportunities remain to reduce loss-making transactions and strengthen underperforming segments.

The findings and recommendations generated through this analysis provide valuable insights that can support strategic decision-making, revenue growth, and long-term business sustainability.